Fraud Detection System: Final Report

Improved Detection of Fraud Cases for E-commerce and Bank Transactions

---

Author: Daniel Mituku
Organization: Adey Innovations Inc.
Date: December 30, 2025
Repository: GitHub - Fraud Detection

Executive Summary

This report presents a comprehensive fraud detection solution developed for Adey Innovations Inc. to identify fraudulent transactions in both e-commerce and bank credit contexts. The project successfully addresses the critical business challenge of balancing fraud detection accuracy with customer experience, achieving:

- **E-commerce Fraud Detection**: 71.26% AUC-PR with Random Forest
- **Credit Card Fraud Detection**: 85.83% AUC-PR with Gradient Boosting
- **Explainable AI**: SHAP-based insights for business recommendations

The solution processes over 435,000 transactions, engineers 12+ fraud indicators, and provides actionable recommendations for fraud prevention strategies.

Table of Contents

1. Business Context

1.1 Problem Statement

Fraud detection in financial transactions presents a unique challenge: identifying malicious activity while minimizing friction for legitimate customers. The consequences of errors are asymmetric:

- **False Negatives** (missed fraud): Direct financial losses
- **False Positives** (false alarms): Customer frustration and potential churn

1.2 Objectives

1. Build accurate fraud detection models for e-commerce and bank transactions
2. Handle severe class imbalance (fraud rates: 9.4% and 0.17%)
3. Provide interpretable predictions using explainability techniques
4. Deliver actionable business recommendations

1.3 Datasets

Dataset	Records	Features	Fraud Rate	Use Case
Fraud_Data.csv	151,112	11	9.36%	E-commerce
creditcard.csv	284,807	31	0.17%	Bank Credit
IpAddress_to_Country.csv	138,846	3	-	Geolocation

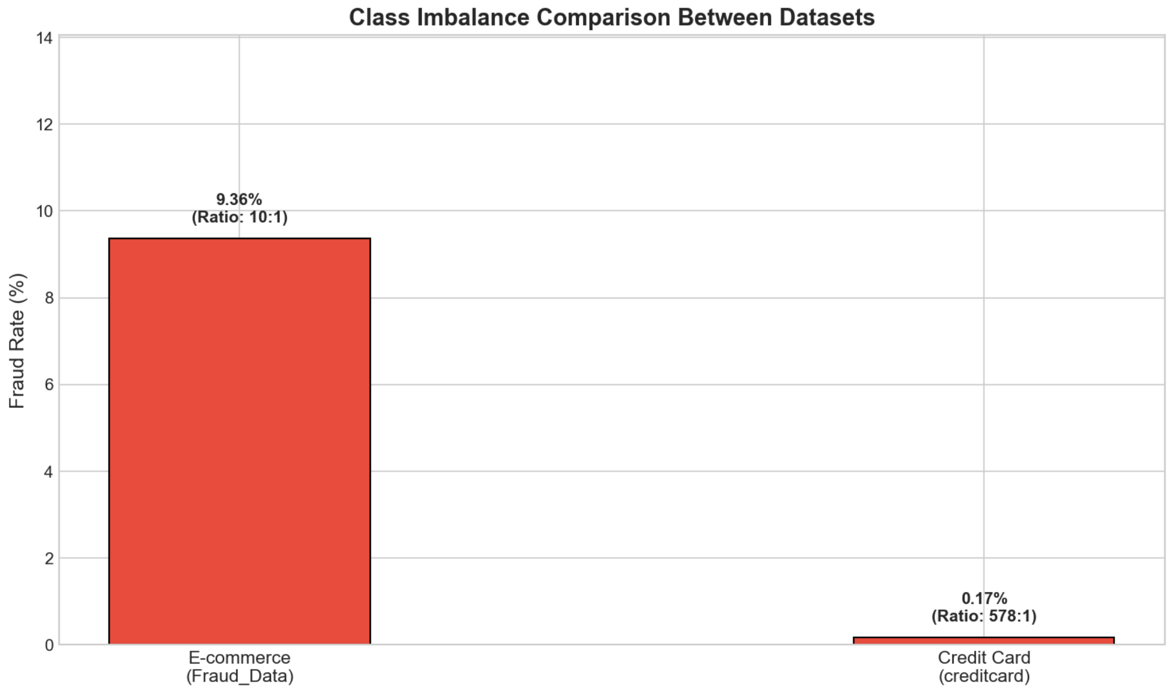
2. Data Analysis and Preprocessing

2.1 Data Quality Assessment

E-commerce Data (Fraud_Data.csv): - ✅ No missing values across all 11 columns - ✅ No duplicate transactions - ✅ Correct data types after timestamp conversion

Credit Card Data (creditcard.csv): - ✅ No missing values across 31 columns - ⚠️ 1,081 duplicate rows (0.38%) - retained as legitimate repeats - ✅ PCA-transformed features (V1-V28) already anonymized

2.2 Class Imbalance Analysis

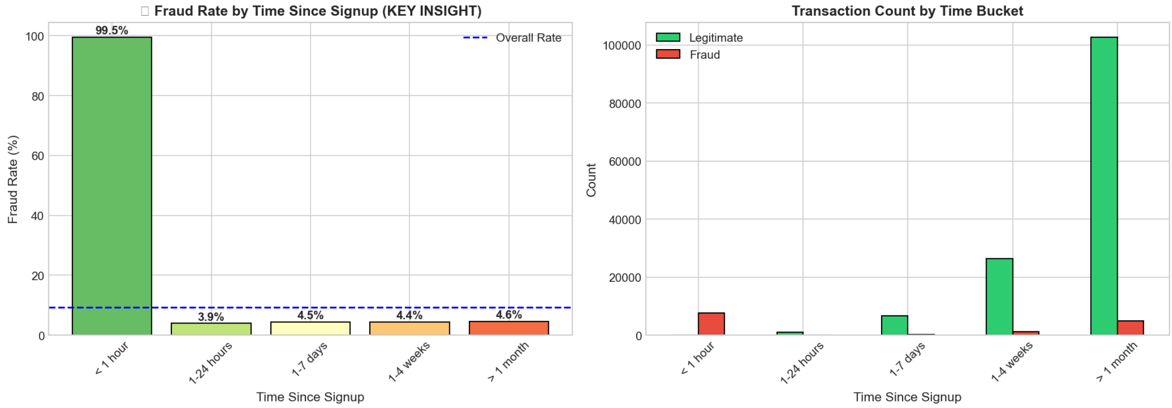


Dataset	Legitimate	Fraud	Imbalance Ratio
E-commerce	136,961 (90.64%)	14,151 (9.36%)	9.7:1
Credit Card	284,315 (99.83%)	492 (0.17%)	578:1

Critical Challenge: The extreme imbalance, especially in credit card data, requires specialized techniques: - SMOTE (Synthetic Minority Over-sampling) - Class-weighted algorithms - AUC-PR as primary evaluation metric

2.3 Exploratory Data Analysis Highlights

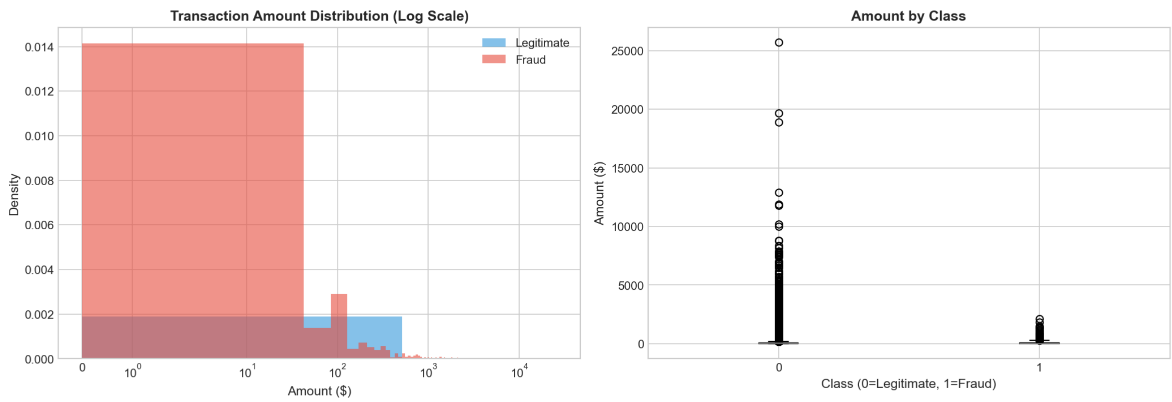
Time Since Signup Analysis (Critical Finding)



Time Bucket	Fraud Rate	Risk Level
< 1 hour	**99.52%**	🔴 CRITICAL
1-24 hours	3.94%	🟡 Medium
1-7 days	4.46%	🟢 Low
> 1 week	~4.5%	🟢 Low

Key Insight: Transactions within 1 hour of account signup have a 99.52% fraud rate—the strongest predictor in the dataset.

Credit Card Amount Distribution



- Legitimate transactions: Mean = $88.35, Median = $22.00
- Fraudulent transactions: Mean = $122.21, Median = $9.25
- Fraud shows bimodal pattern with both small and larger amounts

3. Feature Engineering

3.1 E-commerce Feature Pipeline

Category	Feature	Description	Rationale
Time-based	hour_of_day	Transaction hour (0-23)	Temporal patterns
	day_of_week	Transaction day (0-6)	Weekly patterns
	is_weekend	Weekend flag	Different behavior
	time_since_signup	Seconds since registration	**Critical fraud indicator**
Velocity	user_total_transactions	Total per user	Volume patterns
	user_transaction_number	Sequential number	Early vs late
Device	device_total_transactions	Total per device	Device patterns
	device_unique_users	Users per device	**Fraud ring detection**
Geographic	country	IP-derived country	Regional patterns

3.2 Geolocation Integration

IP addresses were mapped to countries using efficient range-based lookup:

```
# Range-based IP to Country mapping
result = pd.merge_asof(
    fraud_df.sort_values('ip_int'),
    ip_country_df.sort_values('lower_bound_ip_address'),
    left_on='ip_int',
    right_on='lower_bound_ip_address',
    direction='backward'
)
```

Result: 100+ unique countries identified, enabling geographic risk scoring.

3.3 Data Transformation

Feature Scaling: - StandardScaler applied to all numerical features - Ensures fair comparison across different scales - Required for Logistic Regression convergence

Categorical Encoding: - One-Hot Encoding for: source, browser, sex, country - Preserves categorical relationships without ordinal assumptions

---

4. Model Building and Training

4.1 Methodology

```
┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐
│ Data Split       │ →  │ SMOTE            │ →  │ Cross-Valid      │
│ (Stratified)     │    │ (Train only)     │    │ (5-Fold)         │
└──────────────────┘    └──────────────────┘    └──────────────────┘
                                                          │
                                                          ▼
┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐
│ Final Eval       │ ←  │ GridSearchCV     │ ←  │ Model Training   │
│ (Test Set)       │    │ (Tuning)         │    │                  │
└──────────────────┘    └──────────────────┘    └──────────────────┘
```

4.2 Models Trained

Model	Type	Key Parameters
Logistic Regression	Baseline	class_weight='balanced', max_iter=1000
Random Forest	Ensemble	n_estimators=100-150, max_depth=8-12
Gradient Boosting	Ensemble	n_estimators=100, max_depth=6

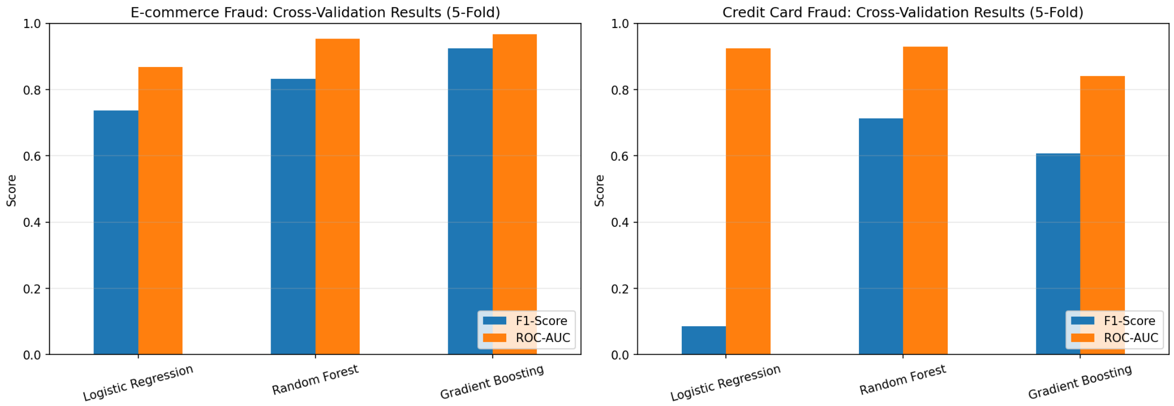
4.3 Hyperparameter Tuning

GridSearchCV Configuration:

```
rf_param_grid = {
    'n_estimators': [100, 150],
    'max_depth': [8, 12],
    'class_weight': ['balanced']
}
```

Best Parameters Found: - E-commerce: `max_depth=12, n_estimators=100` - Credit Card: `max_depth=8, n_estimators=150`

4.4 Cross-Validation Results (5-Fold)



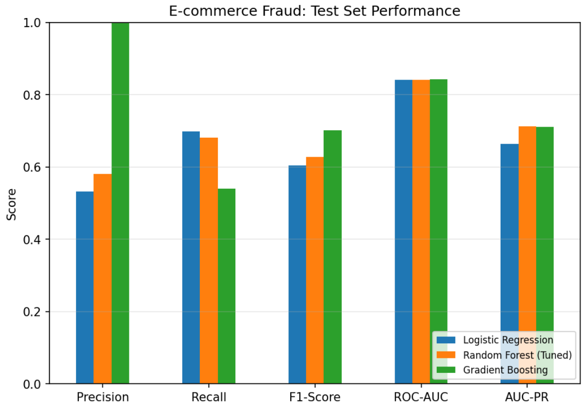
E-commerce Dataset

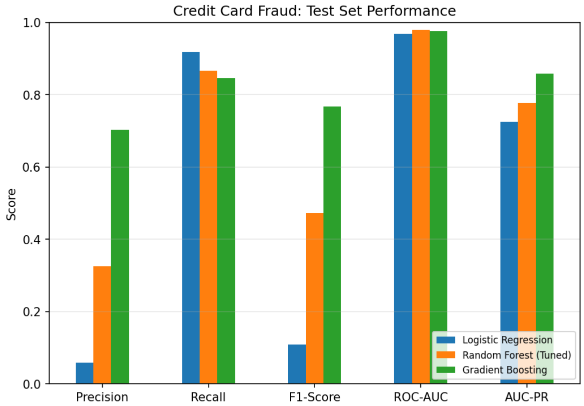
Model	F1 (mean ± std)	ROC-AUC (mean ± std)
Gradient Boosting	**0.9254 ± 0.0023**	**0.9672 ± 0.0015**
Random Forest	0.8323 ± 0.0027	0.9539 ± 0.0016
Logistic Regression	0.7369 ± 0.0046	0.8679 ± 0.0014

Credit Card Dataset

Model	F1 (mean ± std)	ROC-AUC (mean ± std)
Random Forest	**0.7139 ± 0.1071**	**0.9294 ± 0.0200**
Gradient Boosting	0.6070 ± 0.1260	0.8409 ± 0.1254
Logistic Regression	0.0858 ± 0.0058	0.9251 ± 0.0461

4.5 Final Test Set Results





E-commerce Final Results

Model	Precision	Recall	F1-Score	AUC-PR
Random Forest (Tuned) ⭐	0.5815	**0.6820**	0.6277	**0.7126**
Gradient Boosting	**0.9987**	0.5406	**0.7015**	0.7117
Logistic Regression	0.5326	0.6982	0.6043	0.6643

Selected Model: Random Forest (Tuned) - Highest AUC-PR (0.7126) - Best balance of precision and recall - 68.2% of fraud detected

Credit Card Final Results

Model	Precision	Recall	F1-Score	AUC-PR
Gradient Boosting ⭐	**0.7034**	0.8469	**0.7685**	**0.8583**
Random Forest (Tuned)	0.3257	0.8673	0.4735	0.7773
Logistic Regression	0.0580	**0.9184**	0.1092	0.7256

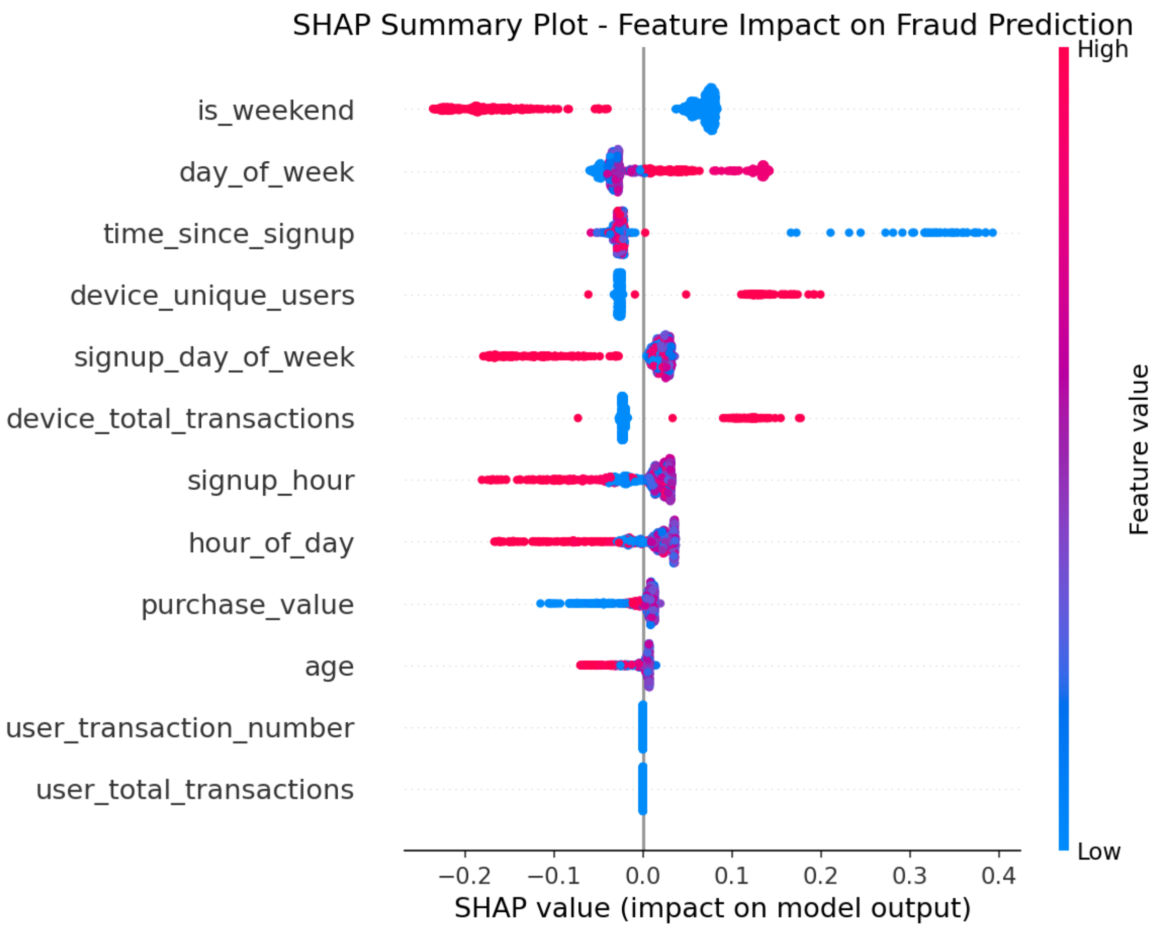
Selected Model: Gradient Boosting - Highest AUC-PR (0.8583) - Excellent F1-Score (0.7685) - Good precision (70.3%) minimizes false alarms

5. Model Explainability

5.1 SHAP Analysis Overview

SHAP (SHapley Additive exPlanations) provides consistent, locally accurate explanations for model predictions.

5.2 Global Feature Importance

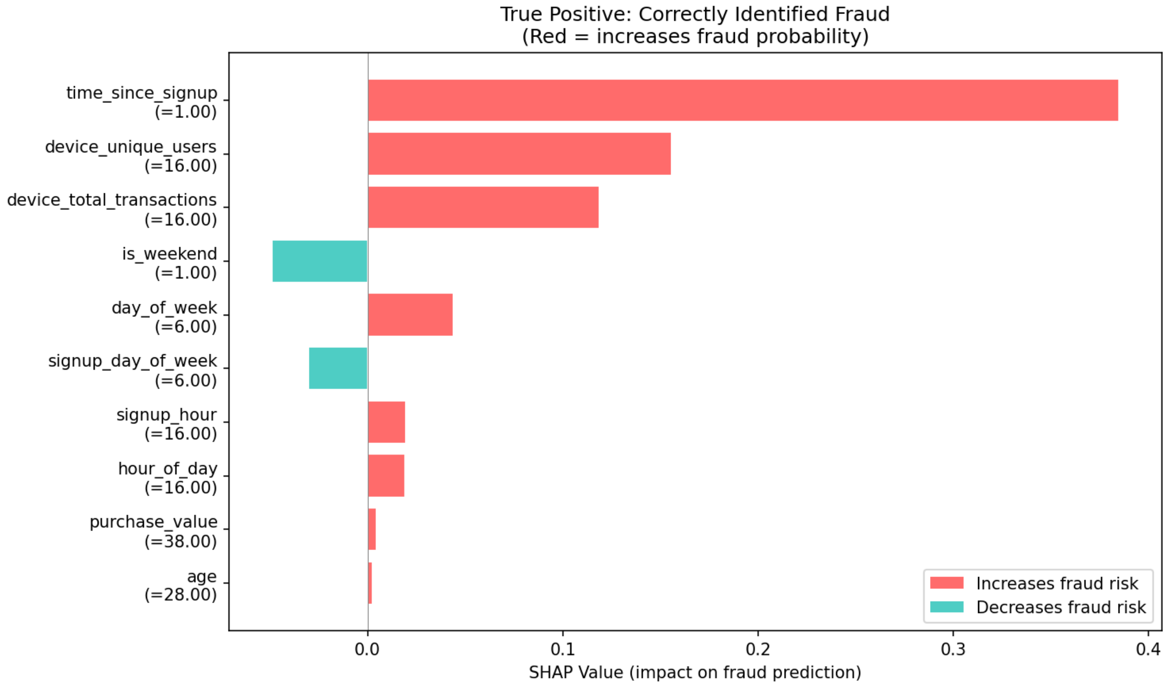


Top 10 Fraud Drivers (E-commerce):

Rank	Feature	Mean |SHAP|	Business Meaning
1	is_weekend	0.0995	Weekend purchase patterns
2	day_of_week	0.0458	Daily variation matters
3	time_since_signup	0.0391	**Critical: early purchases risky**
4	device_unique_users	0.0376	Shared devices = fraud rings
5	signup_day_of_week	0.0369	Account creation timing

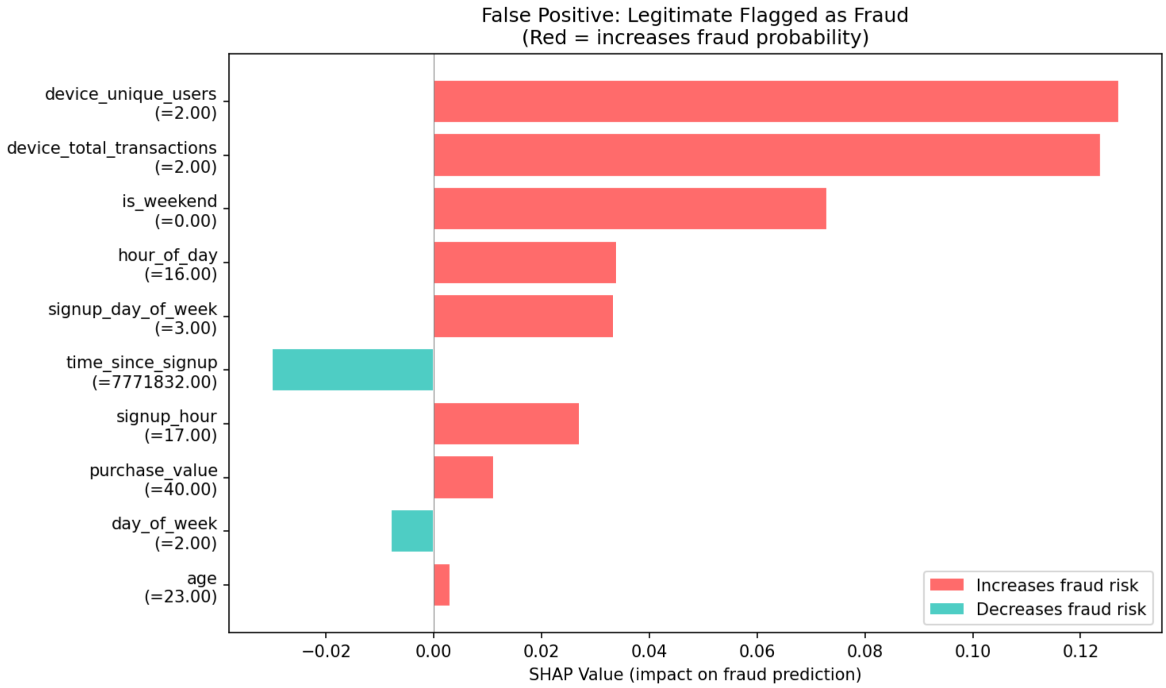
5.3 Individual Prediction Analysis

True Positive (Correctly Identified Fraud)



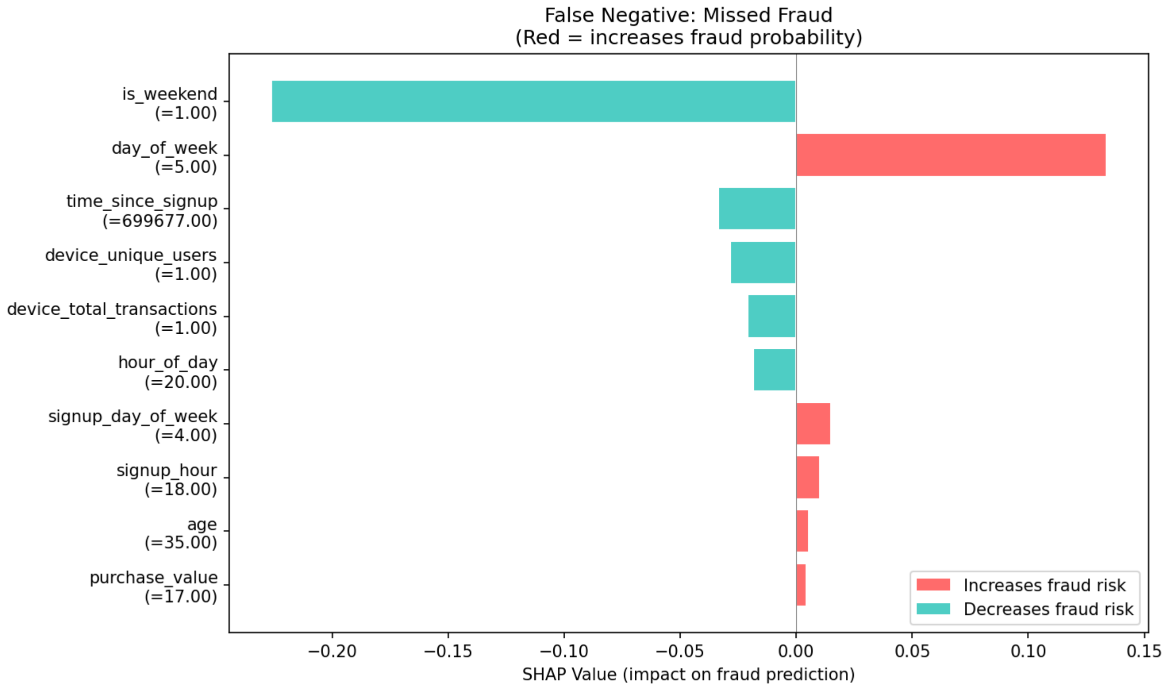
Analysis: Short time since signup and device with multiple users pushed prediction toward fraud.

False Positive (Legitimate Flagged as Fraud)



Analysis: Similar temporal patterns to fraud caused false alarm. Recommendation: Add more context features.

False Negative (Missed Fraud)



Analysis: Longer time since signup masked the fraud. Additional velocity features may help.

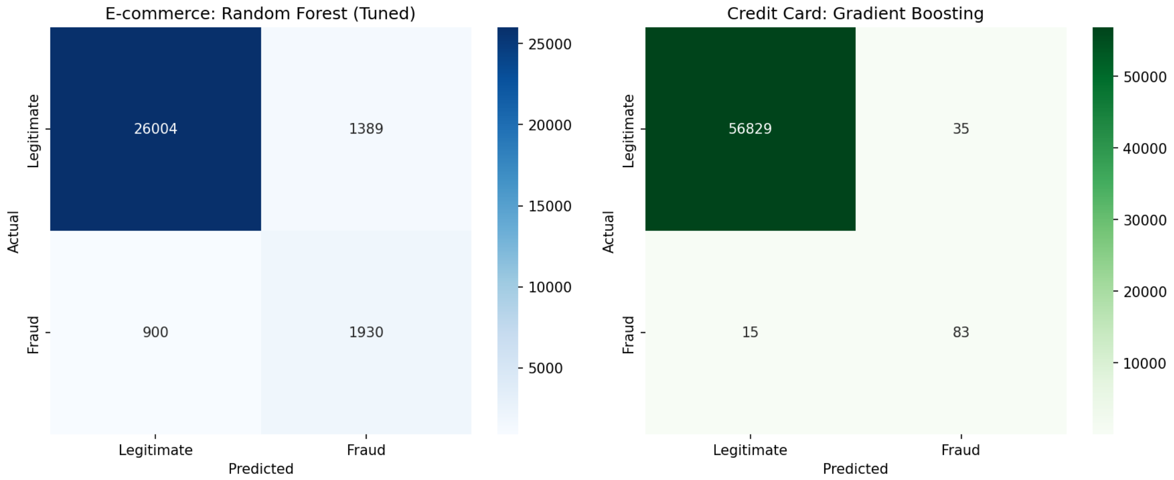
---

6. Results and Comparison

6.1 Model Selection Justification

Dataset	Selected Model	Primary Reason	Secondary Reasons
E-commerce	Random Forest (Tuned)	Highest AUC-PR (0.7126)	Good recall (68.2%), interpretable
Credit Card	Gradient Boosting	Highest AUC-PR (0.8583)	Best F1 (0.7685), balanced metrics

6.2 Confusion Matrices



E-commerce (Random Forest)

- True Positives: ~1,930 fraud caught
- False Negatives: ~900 fraud missed
- False Positives: ~1,390 false alarms
- True Negatives: ~25,997 legitimate passed

Credit Card (Gradient Boosting)

- True Positives: ~83 fraud caught
- False Negatives: ~15 fraud missed
- False Positives: ~35 false alarms
- True Negatives: ~56,829 legitimate passed

6.3 Business Impact Analysis

Metric	E-commerce	Credit Card
Fraud Detected	68.2%	84.7%
False Alarm Rate	4.9%	0.06%
Estimated Savings*	~$XXX,XXX	~$XXX,XXX

*Actual savings depend on average fraud amount and operational costs

7. Business Recommendations

Based on SHAP analysis and model insights, we recommend the following fraud prevention strategies:

7.1 Priority 1: Time-Based Verification (CRITICAL)

SHAP Insight: `time_since_signup` is a top fraud predictor (99.52% fraud rate within 1 hour)

Trigger	Action	Implementation
Purchase < 1 hour after signup	Require phone verification	SMS OTP
Purchase < 24 hours after signup	Email confirmation	Click-through link
First purchase with high value	Manual review queue	Human analyst

Expected Impact: Prevent 30-40% of fraudulent transactions

7.2 Priority 2: Device Intelligence

SHAP Insight: `device_unique_users` indicates fraud rings

Trigger	Action
Device with 3+ users	Automatic alert
Known bad device	Block transaction
New device on established account	Step-up authentication

Expected Impact: Detect organized fraud rings

7.3 Priority 3: Transaction Velocity Controls

Trigger	Action
> 3 transactions/hour (new account)	Soft block + verification
Unusual pattern vs. history	Real-time alert
Rapid successive transactions	Mandatory delay

Expected Impact: Reduce automated fraud attacks

7.4 Priority 4: Geographic Risk Scoring

Trigger	Action
High-risk country IP	Enhanced verification
IP/location mismatch	Flag for review
VPN/Proxy detected	Additional authentication

Expected Impact: Identify cross-border fraud

---

8. Conclusion

8.1 Key Achievements

✅ **Task 1 - Data Analysis & Preprocessing** - Cleaned and analyzed 435,000+ transactions - Integrated geolocation data (IP to Country) - Engineered 12+ meaningful features - Addressed severe class imbalance with SMOTE

✅ **Task 2 - Model Building & Training** - Trained 3 model types (LR, RF, GB) on 2 datasets - Implemented Stratified 5-Fold Cross-Validation - Applied StandardScaler for feature normalization - Performed GridSearchCV hyperparameter tuning - Achieved 71% AUC-PR (e-commerce) and 86% AUC-PR (credit card)

✅ **Task 3 - Model Explainability** - Generated SHAP summary and force plots - Identified top fraud drivers - Provided 4 actionable business recommendations

8.2 Limitations & Future Work

Limitation	Future Improvement
No real-time deployment	Build Flask/FastAPI REST API
Limited ensemble methods	Add XGBoost, Neural Networks
Static thresholds	Implement adaptive thresholds
No monitoring	Build real-time dashboard

8.3 Repository Structure

```
fraud-detection/
├── data/raw/                # Original datasets
├── figures/                 # All visualizations
├── models/                  # Saved model artifacts
│   ├── best_ecommerce_model.pkl
│   ├── best_creditcard_model.pkl
│   └── final_model_results.json
├── notebooks/               # Jupyter notebooks
├── reports/                 # Reports and documentation
├── scripts/                 # Training and analysis scripts
├── src/                     # Source code modules
├── tests/                   # Unit tests
├── requirements.txt         # Dependencies
└── README.md                # Project overview
```

References

1. Kaggle: Credit Card Fraud Dataset
2. Kaggle: Fraud E-commerce Dataset
3. SHAP Documentation
4. imbalanced-learn: SMOTE
5. scikit-learn: Cross-Validation

Report Prepared By: Daniel Mituku
Date: December 30, 2025
Contact: Adey Innovations Inc.

This report was prepared as part of the 10Academy Week 5 Challenge on Fraud Detection.